

Answer Key - CSE-476: Introduction to Natural Language Processing
Quiz 7

(Answers are ordered exactly as in the question paper.)

Multiple Choice Questions

Q1. **Answer:** B

Options: A) It cannot process text input; B) It struggles with noisy or irrelevant retrieved data; C) It cannot use neural networks; D) It only works for images

Note: Correct option: B - A core weakness of vanilla RAG is that wrong or noisy context still gets fed to the generator, which can mislead the model even when retrieval “works.”

Q2. **Answer:** B

Options: A) Relevant documents are evenly distributed across rankings; B) The first relevant document appears early, but others are ranked poorly; C) All relevant documents appear at the top; D) No relevant documents are retrieved

Note: Correct option: B - MRR rewards finding one relevant item high early; MAP averages precision over all relevant items, so if later relevant docs rank badly, MAP can stay low while MRR stays high.

Q3. **Answer:** A

Options: A) It accounts for ranking positions of all relevant documents; B) It only considers the first relevant document; C) It ignores irrelevant documents; D) It requires fewer computations

Note: Correct option: A - MAP incorporates where every relevant document appears in the ranked list, not just a single cutoff like Precision@ k .

Q4. **Answer:** B

Options: A) To enforce similarity between unrelated queries and documents; B) To push irrelevant documents away in embedding space; C) To reduce model complexity; D) To increase dataset size

Note: Correct option: B - Contrastive training in DPR uses negatives so the model learns to separate query–passage pairs that should not match (large margin in embedding space).

Q5. **Answer:** A

Options: A) DPR requires labeled training data and may not generalize well; B) BM25 uses neural embeddings; C) BM25 ignores word frequencies; D) DPR does not use queries

Note: Correct option: A - Dense retrievers depend on domain-matched training; BM25 is unsupervised lexical scoring and can be more stable out-of-domain when DPR has domain shift.

Q6. **Answer:** A

Options: A) To produce diverse reasoning paths; B) To reduce computation time; C)

To eliminate randomness; D) To ensure deterministic outputs

Note: Correct option: A - Self-consistency samples multiple reasoning chains; non-zero temperature yields diversity so different paths can be aggregated (e.g., majority vote).

Q7. **Answer:** A

Options: A) The model may fail to learn fine-grained differences; B) The model learns stronger distinctions; C) The training becomes unstable; D) The embeddings collapse

Note: Correct option: A - If negatives are trivially dissimilar, the task is too easy and the encoder may not learn subtle discrimination between hard/confusable passages.

Q8. **Answer:** D

Options: A) Improved model confidence; B) Reduced inference time; C) Automatic conflict resolution; D) Increased hallucination risk

Note: Correct option: D - Conflicting evidence in context can confuse the model or encourage it to blend or invent a unified answer, raising hallucination risk without explicit conflict handling.

Q9. **Answer:** B

Options: A) It reduces randomness; B) It allows more intermediate reasoning steps; C) It eliminates errors; D) It speeds up inference

Note: Correct option: B - Longer CoT gives room for explicit intermediate steps, which can improve multi-step reasoning when the model uses them correctly (length alone does not guarantee correctness).

Q10. **Answer:** C

Options: A) Train larger models; B) Increase dataset size; C) Extend reasoning at inference time; D) Reduce token usage

Note: Correct option: C - Test-time scaling allocates more compute or search (e.g., more samples, longer chains, verifier steps) at inference rather than only scaling training size or parameters.

End of Answer Key